

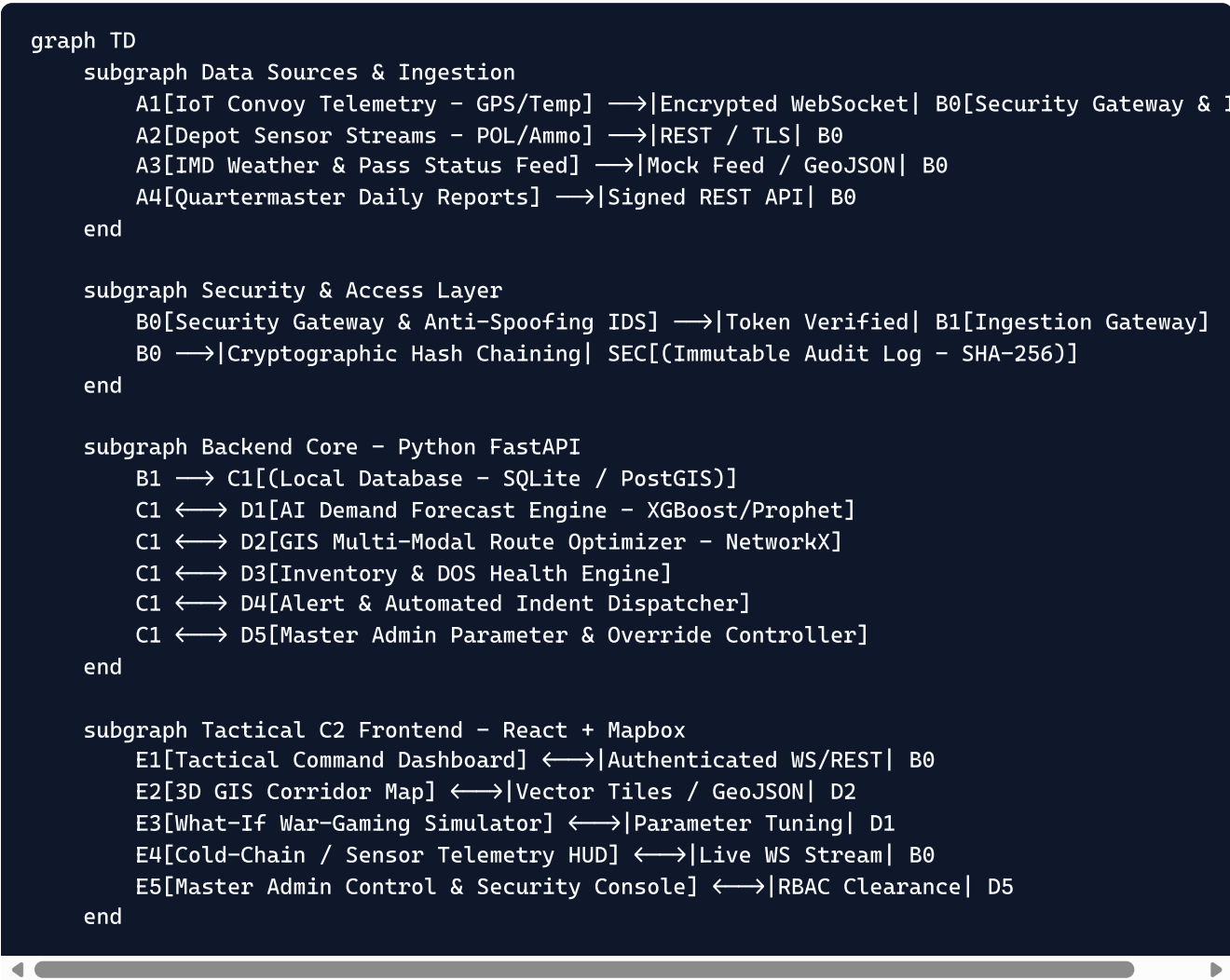
System Architecture Document

System Name: Project Rudra-Logistics

Version: 1.0.0
Target Deployment: Air-gapped Tactical Operational Center (TOC) / Forward Logistics Depots

1. High-Level Architecture Overview

Project Rudra-Logistics follows a modular, decoupled architecture consisting of an **Ingestion & Telemetry Pipeline**, an **AI/Analytics Core**, a **Spatial/GIS Routing Engine**, and a **Tactical Command Frontend**.



2. Technology Stack Selection Rationale

Component	Technology	Rationale for Military Context
Frontend Framework	React 18 + Vite + TailwindCSS	Fast load times, lightweight client bundle, easy component modularity.
Tactical GIS / Maps	Mapbox GL JS / Leaflet + Deck.gl	High-performance client-side vector rendering, offline mbtiles support, 3D terrain elevation.
Backend Framework	Python 3.11 + FastAPI	Async native (asyncio), high concurrency for telemetry, native integration with AI/ML and geospatial packages.
Database	SQLite with Spatialite / PostgreSQL with PostGIS	Standalone offline single-binary portability for prototype; enterprise PostGIS for production.
AI / Machine Learning	Scikit-learn, XGBoost, Pandas, NumPy	Low memory overhead, instant inference ($\leq 20\text{ms}$), easily retrained on edge servers.
Real-Time Streaming	FastAPI WebSockets + Pydantic	Sub-second bi-directional telemetry transmission without cloud broker overhead.

3. Database Schema Design (Entity Relationship Model)

```
erDiagram
    MILITARY_NODE ||--o{ INVENTORY_STOCK : stores
    MILITARY_NODE ||--o{ PASS_STATUS : connects_to
    MILITARY_NODE ||--o{ CONVOY : originates_or_terminates
    INVENTORY_STOCK }o--|| SUPPLY_ITEM : categorizes
    CONVOY ||--o{ TELEMETRY_LOG : emits
    MILITARY_NODE ||--o{ DEMAND_FORECAST : has

    MILITARY_NODE {
        string node_id PK "e.g., NODE_LEH_DEPOT"
        string name "Leh Central Logistics Depot"
        string node_type "BASE_DEPOT | TRANSIT_DEPOT | FORWARD_POST"
        float latitude
        float longitude
        float altitude_feet
        int troop_count
        string threat_level "PEACE | HEIGHTENED | ACTIVE"
    }

    SUPPLY_ITEM {
        string item_id PK "e.g., ITEM_POL_KEROSENE"
        string name "Kerosene Heating Grade"
        string supply_class "CLASS_I | CLASS_III | CLASS_V | CLASS_VIII"
        string unit_of_measure "LITERS | PACKETS | ROUNDS | UNITS"
        float base_consumption_per_soldier_day
        boolean is_cold_chain_required
        float min_safe_temp
        float max_safe_temp
    }

    INVENTORY_STOCK {
        string stock_id PK
        string node_id FK
        string item_id FK
        float current_quantity
        float safety_threshold_qty
        float days_of_supply_dos
        datetime last_updated
    }

    CONVOY {
        string convoy_id PK "e.g., CNV_14C_08"
        string call_sign "STALLION ALPHA"
        string origin_node_id FK
        string destination_node_id FK
        string status "PLANNED | EN_ROUTE | DIVERTED | DELIVERED"
        string assigned_route_id
        int vehicle_count
        float total_payload_tonnes
        datetime dispatch_time
        datetime eta
    }
```

```

TELEMETRY_LOG {
    int log_id PK
    string convoy_id FK
    float current_lat
    float current_lon
    float speed_kmh
    float fuel_level_pct
    float cargo_temperature_celsius
    boolean cold_chain_breach
    datetime timestamp
}

USER_ACCOUNT {
    string user_id PK
    string service_number "ARMY_OFFICER_ID"
    string rank_and_name "Lt Col A K Sharma"
    string role "ADMIN | COMMANDER | LOGISTICS_OFFICER | QUARTERMASTER"
    string password_hash "Argon2id"
    string clearance_level "SECRET | TOP_SECRET"
    boolean is_active
    datetime last_login
}

AUDIT_LOG_ENTRY {
    int audit_id PK
    string action_type "PARAM_OVERRIDE | PASS_TOGGLE | DISPATCH_AUTH | STOCK_ADJUST"
    string executed_by_user_id FK
    string details_json
    string prev_hash "SHA256_CHAIN"
    string current_hash "SHA256_CHAIN"
    datetime timestamp
}

```

4. REST & WebSocket API Contracts

4.1 GET `/api/v1/nodes/health`

Returns all military nodes with calculated Days of Supplies (DOS) and alert status.

```
[
  {
    "node_id": "FOP_SIACHEN_BASE",
    "name": "Siachen Base Camp",
    "altitude_feet": 12000,
    "troop_count": 350,
    "temperature_celsius": -28.5,
    "overall_status": "CRITICAL",
    "inventory": [
      {
        "item_id": "ITEM_POL_KEROSENE",
        "name": "Bukhari Kerosene",
        "current_qty": 4200.0,
        "daily_burn_rate": 1400.0,
        "dos": 3.0,
        "status": "RED"
      },
      {
        "item_id": "ITEM_CLASS_I_RATIONS",
        "name": "High Altitude Rations",
        "current_qty": 6300.0,
        "daily_burn_rate": 350.0,
        "dos": 18.0,
        "status": "GREEN"
      }
    ]
  }
]
```

4.2 POST `/api/v1/routes/optimize`

Calculates optimal path taking active roadblock/hazard states into account.

- Request:

```
json { "origin_node_id": "BASE_LEH_DEPOT", "destination_node_id": "FOP_SIACHEN_BASE",
"blocked_passes": ["PASS_KHARDUNG_LA"], "cargo_priority": "CRITICAL_HEATING" }
```

- Response:

```
json { "route_id": "ROUTE_BYPASS_SHYOK", "route_name": "Axis 2: Leh - Karu - Chang La - Agham
- Nubra - Siachen", "distance_km": 284.5, "estimated_travel_time_hours": 9.2,
"elevation_gain_m": 4200, "hazard_level": "MODERATE", "recommended_convoy_type":
"4x4_ALS_WITH_SNOW_CHAINS", "path_coordinates": [ [34.1526, 77.5771], [34.0253, 77.7423],
[35.3214, 77.2012] ] }
```

4.3 WebSocket `/ws/telemetry`

Live streaming channel updating convoy GPS, vehicle health, and cold-chain temperature alerts every 2 seconds.

4.4 POST `/api/v1/admin/overrides/parameter`

Admin-only endpoint for modifying critical operational burn-rates or safety multipliers.

- **Request:**

```
json { "parameter_name": "BUKHARI_HEATING_BURN_RATE_MULTIPLIER", "target_node_id":  
"FOP_SIACHEN_BASE", "new_value": 3.4, "authorization_token": "JWT_BEARER_SIGNED",  
"justification": "Extreme blizzard forecasted for next 72 hours" }
```

- **Response:**

```
json { "status": "APPLIED", "audit_id": 1042, "sha256_hash":  
"e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855", "message": "Parameter  
updated and appended to immutable audit chain." }
```

4.5 GET `/api/v1/security/audit-chain/verify`

Validates the cryptographic integrity of the entire supply chain audit trail.

- **Response:**

```
json { "chain_length": 842, "tamper_detected": false, "latest_block_hash": "9f83c68a ... 7a2",  
"status": "CRYPTOGRAPHICALLY_VERIFIED" }
```

5. Air-Gapped & Offline Architecture

In military tactical sectors, commercial cloud infrastructure (AWS/GCP/Azure) is **prohibited for classified operations**.

1. **Self-Contained Edge Deployment:** The entire stack (React UI, FastAPI server, SQLite/PostGIS, ML weights) is packaged into a local Docker container or standalone executable.
 2. **Local Mesh Sync:** Transit Depots run local server instances. When a convoy arrives at a transit camp, local WiFi/RFID or tactical UHF radios synchronize transaction logs peer-to-peer.
 3. **No External CDN Dependencies:** All frontend icons, fonts (Inter/JetBrains Mono), and map tiles are bundled locally without fetching from external Google Fonts or CDN URLs.
-

6. Military Cyber Security & Anti-Hacking Framework

```
graph LR
    subgraph Threat_Surface
        T1[Brute Force Login]
        T2[GPS / Sensor Spoofing]
        T3[Insider Data Tampering]
        T4[Injection & Replay Attacks]
    end

    subgraph Defense_&_Countermeasures
        T1 --> D1[Argon2id + Sliding Window Rate Limiter]
        T2 --> D2[Kinematic Heuristic Anti-Spoofing Filter]
        T3 --> D3[SHA-256 Cryptographic Hash Chain]
        T4 --> D4[Pydantic Type Validation + Param SQL]
    end

    subgraph Operational_Outcome
        D1 & D2 & D3 & D4 --> RES[Hardened Tactical C2 System]
    end
```

6.1 Cryptographic Audit Trail (Tamper-Evident Ledger)

To prevent rogue actors or supply diversions from cooking digital inventory logs:

- Every inventory increment, convoy authorization, or master override is structured as a transaction.
- Each transaction stores:

$$\text{Current Hash} = \text{SHA256}(\text{Prev Hash} + \text{Timestamp} + \text{Officer ID} + \text{Action Data})$$

- If an adversary directly edits the local SQLite/Postgres database using external tools, the hash chain breaks instantly, triggering a **CRITICAL SYSTEM COMPROMISE** red lockdown on the Commander's C2 HUD.

6.2 Telemetry Anomaly & Anti-Spoofing Heuristics

Adversary Electronic Warfare (EW) units routinely deploy GPS spoofing near borders:

- **Kinematic Plausibility Check:** If a convoy jumps coordinates at an impossible speed ($> 90 \text{ km/h}$) on mountain dirt passes) or reports negative altitude, the packet is flagged as `SPOOFED_TELEMETRY`.
- The system automatically ignores the spoofed blip and estimates dead reckoning position based on convoy last known speed and route plan.

6.3 Hardened Defense Against OWASP Top 10

- **SQL Injection (SQLi):** 100% prevented via SQLAlchemy ORM parameterized queries; raw SQL queries are strictly prohibited in the codebase.
- **Cross-Site Scripting (XSS):** React auto-escaping + strict Content Security Policy (CSP).

- **Session Hijacking:** JWT tokens stored in memory/HttpOnly secure cookies with short 15-minute expirations and cryptographic signature checks.